

Introduction to Computer Programming (ICP, CCIT4020)

Assignment 3 (Written)

(15%)

(Total Marks: 60)

Answer Book

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Instructions:

- Follow instructions, including those in the question paper and the answer book.
- Print out this answer book** for answering questions with **HANDWRITING**. Use black or dark-blue pen to answer. Make sure answers are well-written. **TYPING / POOR Writing may NOT be marked!!**
 - In case students do NOT have printers, they may use A4-size blank/single-line paper(s) with format similar to this given answer book (e.g. draw proper section lines) to answer questions.
- Properly Scan/take-photo** the finished answer book, **convert it into ONE pdf file**, and **submit the pdf file to SOUL**. Scanned/photo-taken copies (submitted to SOUL) must be **clear enough for printing-out and marking**. You may need to try in different light conditions.

Section A: Multiple Choice Questions (20 marks)

1	2	3?	4	5
C	A	B	C	C
6?	7	8	9	10
D	D	B	B	A

Section B: Short Questions (40 marks)

B1. (10 marks)

a) [3 marks]

i.	**	*	+
ii.	/	*	-
iii.	/	*	+

b) [3 marks] ~~① "11" is invalid because this identifier start from key word "Valid" and have "1"~~

~~② "111" is invalid because this identifier start from a digit "1"~~

③ "i" is valid because this identifier only consist of letters, digits and underscores, and start with character.

④ "ii" is invalid because this identifier have "scores" (-) which is not allow, and it starts with key word ("Valid")

c) [4 marks]

```
for n in range(6, 31, 3):
```

```
    print(n, end = "> ")
```

↑

space line

">" one space

B2. (10 marks)

a) [4 marks]

```
str1 = "c"
```

```
n = 10
```

```
while n >= 2:
```

```
    print(n * str1)
```

```
    n = n - 2
```

b) [6 marks]

~~aStr~~

```
aStr = str(input("Enter a string: "))
```

```
fStr = aStr[:len(aStr)//4]
```

```
eStr = aStr[(len(aStr)*3//4):len(aStr)]
```

```
print(fStr + eStr)
```

B3. (20 marks)

a) [8 marks]

```
def firstEven():  
    mList = [9, 1, 0, 2]  
    i = 0  
    while i < len(mList):  
        if mList[i] % 2 == 0:  
            return mList[i]  
            break  
        else:  
            i += 1
```

~~firstEven()~~

b) [7 marks]

```
def icpTotal():  
    sentence = str(input("Enter a string: "))  
    characters = ["i", "c", "p"]  
    Total = 0  
    for word in sentence:  
        if word in characters:  
            Total += 1  
    print("Total i, c, p in string: ", Total)
```

~~icpTotal()~~

c) [5 marks]

```
def main():  
    print("Testing by <Yenny Man>")  
    print(firstEven())  
    icpTotal()
```

main()

* Use extra pages, if given space is not enough.